Inverse Function Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of inverse function theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AC-04 / Advanced Calculus / Report v1

I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $U \subset \mathbb{R}^n$ be open, $F : U \to \mathbb{R}^n$ be C^1, and $a \in U$ have invertible derivative $L = DF(a)$. There are open neighborhoods V of a and W of $F(a)$ such that $F : V \to W$ is a bijection with C^1 inverse H, and*

$$DH(y) = DF(H(y))^{-1}.$$

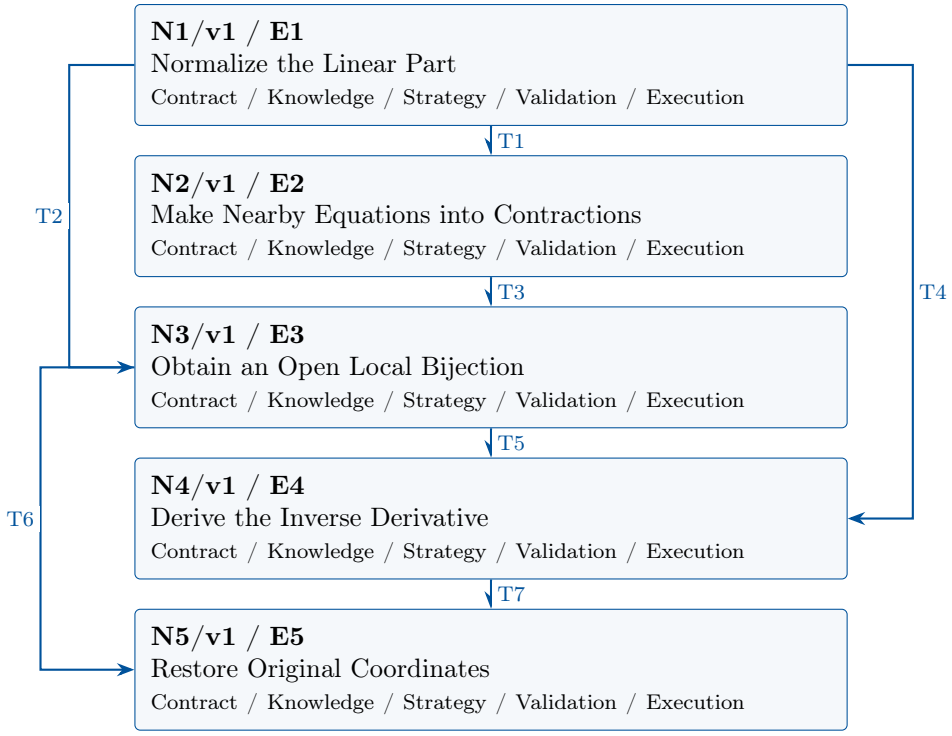
The goal has four obligations: existence of nearby solutions, their local uniqueness, continuous dependence on the target, and differentiability of the inverse. An invertible Jacobian at one point does not assert global bijectivity.

A. Foundations and permitted imports

Euclidean completeness, integration of continuous derivatives along segments, elementary matrix inversion, and the completed Banach fixed-point module. Operator norms are induced by the Euclidean norm. The proof constructs its neighborhoods rather than assuming an inverse already exists.

The segment integral estimate is an application of the one-variable FTC (AC-02) component by component. The Banach contraction theorem is an external Real Analysis import, not an extra node in this subject graph.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. $G(z) = L^{-1}(F(a+z) - F(a))$ has $DG(0) = I$ and $E = G - I$ is $1/2$-Lipschitz on a closed ball of radius ρ.

Receiving activity. N2: Make Nearby Equations into Contractions

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1; completeness of the closed Euclidean ball.

T2: N1/E1/v1 to N3. $G(z) = L^{-1}(F(a+z) - F(a))$ has $DG(0) = I$ and $E = G - I$ is $1/2$-Lipschitz on a closed ball of radius ρ.

Receiving activity. N3: Obtain an Open Local Bijection

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, and the imported Banach contraction theorem.

T3: N2/E2/v1 to N3. For $\|y\| < \rho/2$, $T_y(z) = y - E(z)$ is a contraction mapping $\overline{B}_\rho$ into itself.

Receiving activity. N3: Obtain an Open Local Bijection

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, and the imported Banach contraction theorem.

T4: N1/E1/v1 to N4. $G(z) = L^{-1}(F(a+z) - F(a))$ has $DG(0) = I$ and $E = G - I$ is $1/2$-Lipschitz on a closed ball of radius ρ.

Receiving activity. N4: Derive the Inverse Derivative

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and N3/E3; the Neumann-series inverse bound.

T5: N3/E3/v1 to N4. $G : V_0 \to B_{\rho/2}$ is a bijection with a 2-Lipschitz inverse H_0, where $V_0 = B_\rho \cap G^{-1}(B_{\rho/2})$ is open.

Receiving activity. N4: Derive the Inverse Derivative

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and N3/E3; the Neumann-series inverse bound.

T6: N3/E3/v1 to N5. $G : V_0 \to B_{\rho/2}$ is a bijection with a 2-Lipschitz inverse H_0, where $V_0 = B_\rho \cap G^{-1}(B_{\rho/2})$ is open.

Receiving activity. N5: Restore Original Coordinates

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N3/E3 and N4/E4 with the fixed affine coordinate changes.

T7: N4/E4/v1 to N5. $DH_0(y) = DG(H_0(y))^{-1}$ exists and is continuous.

Receiving activity. N5: Restore Original Coordinates

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N3/E3 and N4/E4 with the fixed affine coordinate changes.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Normalize the Linear Part

Object and role. Normalize the Linear Part. Establish the following local result: $G(z) = L^{-1}(F(a+z) - F(a))$ has $DG(0) = I$ and $E = G - I$ is 1/2-Lipschitz on a closed ball of radius ρ.

Inputs. $F \in C^1(U, \mathbb{R}^n)$, $a \in U$, and $L = DF(a)$ invertible.

C — Contract

Own E1/v1: $G(z) = L^{-1}(F(a+z)-F(a))$ has $DG(0) = I$ and $E = G-I$ is 1/2-Lipschitz on a closed ball of radius ρ.

K — Knowledge

Jacobian and linear inverses

S — Strategy

Make derivative identity at zero

V — Validation

Require invertible L

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Translate a and $F(a)$ to zero and multiply by L^{-1}.

N1.2 Use continuity of DG to choose a closed ball inside the transformed domain with $\|DG-I\| \leq 1/2$.

N1.3 Integrate DE along segments to obtain the Lipschitz estimate.

Export E1/v1. $G(z) = L^{-1}(F(a+z) - F(a))$ has $DG(0) = I$ and $E = G - I$ is 1/2-Lipschitz on a closed ball of radius ρ.

Acceptance gate. Require invertible L Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Require invertible L Recheck dependent nodes N2, N3, N4, N5.

B. N2/v1: Make Nearby Equations into Contractions

Object and role. Make Nearby Equations into Contractions. Establish the following local result: For $\|y\| < \rho/2$, $T_y(z) = y - E(z)$ is a contraction mapping $\overline{B}_\rho$ into itself.

Inputs. N1/E1; completeness of the closed Euclidean ball.

C — Contract

Own E2/v1: For $\|y\| < \rho/2$, $T_y(z) = y - E(z)$ is a contraction mapping $\overline{B}_\rho$ into itself.

K — Knowledge

Derivative continuity and segment estimates

S — Strategy

Recast equation as fixed point

V — Validation

Check complete invariant ball

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Estimate $\|T_y(z) - T_y(w)\| \le \|z - w\|/2$.

N2.2 Use $E(0) = 0$ to prove $\|T_y(z)\| \le \|y\| + \|z\|/2 < \rho$.

N2.3 Check both the uniform contraction constant and the self-map condition before applying Banach.

Export E2/v1. For $\|y\| < \rho/2$, $T_y(z) = y - E(z)$ is a contraction mapping $\overline{B}_\rho$ into itself.

Acceptance gate. Check complete invariant ball Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Check complete invariant ball Recheck dependent nodes N3, N4, N5.

C. N3/v1: Obtain an Open Local Bijection

Object and role. Obtain an Open Local Bijection. Establish the following local result: $G : V_0 \to B_{\rho/2}$ is a bijection with a 2-Lipschitz inverse H_0, where $V_0 = B_\rho \cap G^{-1}(B_{\rho/2})$ is open.

Inputs. N1/E1, N2/E2, and the imported Banach contraction theorem.

C — Contract

Own E3/v1: $G : V_0 \to B_{\rho/2}$ is a bijection with a 2-Lipschitz inverse H_0, where $V_0 = B_\rho \cap G^{-1}(B_{\rho/2})$ is open.

K — Knowledge

Banach existence and uniqueness

S — Strategy

Separate onto and one-to-one

V — Validation

Keep solutions inside open neighborhood

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 For each target, solve $z = T_y(z)$ uniquely on the closed ball.

N3.2 Use the strict self-map bound to place the solution inside the open ball.

N3.3 Prove $\|G(z) - G(w)\| \geq \|z - w\|/2$ and deduce inverse continuity and injectivity.

Export E3/v1. $G : V_0 \to B_{\rho/2}$ is a bijection with a 2-Lipschitz inverse H_0, where $V_0 = B_\rho \cap G^{-1}(B_{\rho/2})$ is open.

Acceptance gate. Keep solutions inside open neighborhood Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Keep solutions inside open neighborhood Recheck dependent nodes N4, N5.

D. N4/v1: Derive the Inverse Derivative

Object and role. Derive the Inverse Derivative. Establish the following local result: $DH_0(y) = DG(H_0(y))^{-1}$ exists and is continuous.

Inputs. N1/E1 and N3/E3; the Neumann-series inverse bound.

C — Contract

Own E4/v1: $DH_0(y) = DG(H_0(y))^{-1}$ exists and is continuous.

K — Knowledge

First-order remainder and matrix bounds

S — Strategy

Solve increment relation

V — Validation

Do not assume differentiability of inverse

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 From $\|DG - I\| \le 1/2$ obtain invertibility and $\|DG^{-1}\| \le 2$.

N4.2 Put $h = H_0(y + k) - H_0(y)$; the Lipschitz bound gives $\|h\| \le 2\|k\|$.

N4.3 Solve $k = DG(H_0(y))h + o(\|h\|)$ for h and then use continuity of matrix inversion.

Export E4/v1. $DH_0(y) = DG(H_0(y))^{-1}$ exists and is continuous.

Acceptance gate. Do not assume differentiability of inverse Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Do not assume differentiability of inverse Recheck dependent nodes N5.

E. N5/v1: Restore Original Coordinates

Object and role. Restore Original Coordinates. Establish the following local result: $F : V \to W$ is a C^1 diffeomorphism and $D(F^{-1})(Y) = DF(F^{-1}(Y))^{-1}$.

Inputs. N3/E3 and N4/E4 with the fixed affine coordinate changes.

C — Contract

Own E5/v1: $F : V \to W$ is a C^1 diffeomorphism and $D(F^{-1})(Y) = DF(F^{-1}(Y))^{-1}$.

K — Knowledge

All local certificates

S — Strategy

Transport inverse and derivative

V — Validation

Restore original coordinates

E — Execution

Carry out the numbered deductions and publish E5/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N5.1 Set $V = a + V_0$ and $W = F(a) + LB_{\rho/2}$.

N5.2 Define $H(Y) = a + H_0(L^{-1}(Y - F(a)))$ and check both inverse identities.

N5.3 Apply the chain rule to transport the derivative formula to the original coordinates.

Export E5/v1. $F : V \to W$ is a C^1 diffeomorphism and $D(F^{-1})(Y) = DF(F^{-1}(Y))^{-1}$.

Acceptance gate. Restore original coordinates Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N5: Restore original coordinates Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Normalize the Linear Part

Set

$$G(z) = L^{-1}\big(F(a+z) - F(a)\big).$$

Then $G(0) = 0$ and $DG(0) = I$. Its domain is an open neighborhood of zero. By continuity of DG choose $\rho > 0$ such that the closed Euclidean ball $\overline{B}_\rho$ lies in that domain and

$$\|DG(z) - I\| \le \tfrac{1}{2} \quad (z \in \overline{B}_\rho).$$

Write $E(z) = G(z) - z$. For x,z in this ball the connecting segment remains inside it, and integration of DE along that segment gives

$$\|E(x) - E(z)\| \le \tfrac{1}{2}\|x - z\|.$$

In particular $E(0) = 0$ and $\|E(z)\| \le \|z\|/2$.

B. N2 — Make Nearby Equations into Contractions

For a normalized target y with $\|y\| < \rho/2$, define

$$T_y(z) = y - E(z) = z - G(z) + y.$$

Its fixed points are exactly the solutions $G(z) = y$. The previous estimate gives contraction factor $1/2$. Also

$$\|T_y(z)\| \le \|y\| + \tfrac{1}{2}\rho < \rho$$

for z in the closed ball. Thus T_y maps a complete ball into itself. Both conditions are required to import Banach's theorem.

C. N3 — Obtain an Open Local Bijection

Banach gives exactly one solution in the closed ball, and the strict self-map bound places it in the open ball. Separately, for x,z in the closed ball,

$$\|G(x) - G(z)\| \ge \|x - z\| - \|E(x) - E(z)\| \ge \tfrac{1}{2}\|x - z\|.$$

Hence G is injective on the ball. Put

$$W_0 = B_{\rho/2}, \qquad V_0 = B_\rho \cap G^{-1}(W_0).$$

The set V_0 is open, contains zero, and G maps it bijectively onto W_0. Let H_0 be its inverse. Applying the lower bound to $H_0(y), H_0(y')$ gives

$$\|H_0(y) - H_0(y')\| \leq 2\|y - y'\|.$$

So the inverse is Lipschitz and in particular continuous. This does not yet establish its differentiability.

D. N4 — Derive the Inverse Derivative

Every $DG(x)$ on the ball is invertible: $\|DG(x) - I\| \leq 1/2$ permits the convergent matrix geometric series for its inverse, with inverse norm at most 2.

Fix y in W_0 and $x = H_0(y)$. For small k with y+k in W_0, set $h = H_0(y + k) - H_0(y)$. The Lipschitz bound gives $\|h\| \leq 2\|k\|$. Differentiability of G at x gives

$$k = G(x + h) - G(x) = DG(x)h + r(h), \qquad \|r(h)\| = o(\|h\|).$$

Therefore

$$h = DG(x)^{-1}k - DG(x)^{-1}r(h) = DG(x)^{-1}k + o(\|k\|).$$

The remainder conversion is justified by $\|h\| \leq 2\|k\|$ and the bounded inverse matrix. Thus $DH_0(y) = DG(H_0(y))^{-1}$. This derivative is continuous because H_0 and DG are continuous and matrix inversion is continuous on invertible matrices. Hence H_0 is C^1.

E. N5 — Restore Original Coordinates

Choose $V = a + V_0$ and $W = F(a) + L\ W_0$. These are open neighborhoods. Their inverse is

$$H(Y) = a + H_0\big(L^{-1}(Y - F(a))\big).$$

The chain rule gives $DH(Y) = DF(H(Y))^{-1}$. All four obligations are now proved, so the original-coordinate theorem follows. □

□

V. WORKED ORCHESTRATION

Let $F(x, y) = (x + y^2, y + x^2)$ at $a = (0, 0)$. Its derivative at zero is I. On the closed Euclidean ball of radius $\rho = 1/4$,

$$DF - I = \begin{pmatrix} 0 & 2y \\ 2x & 0 \end{pmatrix}, \qquad \|DF - I\| = 2\max(|x|, |y|) \leq \tfrac{1}{2}.$$

Thus every target of norm less than $1/8$ has exactly one preimage in the chosen local neighborhood. The fixed-point update is

$$(x, y) \longmapsto (u - y^2, v - x^2).$$

For target $(u, v) = (1/20, 1/20)$, symmetry gives $x = y = s$, where

$$s + s^2 = \tfrac{1}{20}, \qquad s = \frac{\sqrt{6/5} - 1}{2} \approx 0.0477225575.$$

The negative quadratic root is outside the local ball and does not contradict local uniqueness. At this preimage,

$$DH = \frac{1}{1 - 4s^2} \begin{pmatrix} 1 & -2s \\ -2s & 1 \end{pmatrix}.$$

The Banach module also supplies an error bound for the numerical iteration, using $q = 1/2$.

A. Boundary and hypothesis audit

At $F(x) = x^3$ near zero, the inverse exists as a continuous map but fails the claimed differentiability; $DF(0)$ is not invertible. Polar coordinates are locally invertible away from $r = 0$ but remain globally periodic in angle. These distinguish failure of a sufficient hypothesis from impossibility of every inverse.

Export the normalized contraction construction, local inverse, Lipschitz bound, and C^1 derivative formula. The example explicitly demonstrates a theorem module imported from another subject's graph.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

- **Interaction**

  N1 → N2

  Certificate

  Normalized map and uniform derivative bound on the closed ball

- **Interaction**

  N2 → N3

  Certificate

Complete self-map, contraction 1/2, target-radius restriction

- **Interaction**

  Banach → N3

  Certificate

  Unique fixed point for each admissible target

- **Interaction**

  N3 → N4

  Certificate

  Local inverse and Lipschitz factor 2

- **Interaction**

  N4 → N5

  Certificate

  C^1 inverse with proved derivative formula

- **Interaction**

  N5 → caller

  Certificate

  Open neighborhoods in original coordinates

A failed branch selects a large target but reuses the small-ball certificate. For the example, a target with norm 1 is not covered by the radius-1/8 guarantee.

- **Node:** N2
- **State:** STALLED
- **Failing component:** Validation
- **Failed hypothesis:** target norm $<$ rho/2
- **Observed:** target norm 1, $\rho = 1/4$
- **Repair target:** target/domain contract
- **Blocked export:** small-ball existence certificate

Knowledge retrieves the radius relation. Validation retains the target mismatch. Strategy either restricts the target or develops a new valid neighborhood. Execution rechecks the self-map estimate. Contract publishes only the result supported by that estimate; failure of this certificate alone is not proof that a large-target solution does not exist.

B. Hypothesis boundary

At $F(x) = x^3$ near zero, the inverse exists as a continuous map but fails the claimed differentiability; $DF(0)$ is not invertible. Polar coordinates are locally invertible away from $r = 0$ but remain globally periodic in angle. These distinguish failure of a sufficient hypothesis from impossibility of every inverse.

Export the normalized contraction construction, local inverse, Lipschitz bound, and C^1 derivative formula. The example explicitly demonstrates a theorem module imported from another subject's graph.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. University of Pennsylvania: contraction-based proof of the inverse function theorem.